

DOLLY POS • MASTER SYSTEM DICTIONARY

Complete Reference Guide for Every .BAT, .PS1, .EXE File & Script in the Dolly POS Ecosystem

Store Name: Dolly Toys & Kids Wear, Dhule | **Software Edition:** Dolly POS v1.3.0 (2026 Production Edition)
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Purpose of this Document: As Dolly POS evolved into an enterprise-grade retail software with PostgreSQL, automated background synchronization, disaster recovery, and dual-mode testing, multiple specialized scripts (.bat, .ps1, .exe) were created. This dictionary serves as your definitive 1-stop reference: it explains exactly **what every file is, where it is located, when to use it, and step-by-step instructions on how to use it.**

Quick Reference Index by Task

What do you want to do?	Recommended Tool / File	Execution Time
Start normal billing at counter	Dolly POS Desktop Icon or DollyPOS_Launcher.bat	< 1 second
Preview / test beta code changes	DollyPOS_Beta_Launcher.bat	2 seconds
PostgreSQL service stopped / dead battery	Heal_and_Restart_PostgreSQL.bat	3 seconds
Switch to another database (e.g. database1)	Change_Database_Connection.bat	5 seconds
Install Dolly POS on a brand new laptop	Setup_PostgreSQL_Database.bat then DollyPOS_Setup_v1.0.0.exe	2 minutes
Compile new source code into final .exe	Update_Production_App.bat	45 seconds
Take manual emergency database backup	backup_db.bat	1 second

1. Everyday Application Launchers

[DAILY PRODUCTION] DollyPOS_Launcher.bat

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\DollyPOS_Launcher.bat

When to Use: Every day at the shop counter when opening the cash desk.

What it Does: Launches the compiled native standalone Dolly POS executable (DollyPOS.exe) in clean production mode with zero terminal windows.

How to Use / Steps:

- Step 1:** Double-click DollyPOS_Launcher.bat (or double-click the 'Dolly POS' Desktop shortcut).
- Step 2:** The login screen appears in < 1 second. Log in with Owner (admin) or Staff (staff).

Step 3: Begin scanning barcodes and billing customers immediately.

[BETA / TESTING] DollyPOS_Beta_Launcher.bat

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\DollyPOS_Beta_Launcher.bat

When to Use: When you ask for code changes, UI tweaks, or new features and want to live-test them before updating the production desktop app.

What it Does: Starts both the FastAPI development backend server and the Vite React frontend with instant hot-reload, opening your default web browser at <http://localhost:5173> automatically.

How to Use / Steps:

Step 1: Double-click DollyPOS_Beta_Launcher.bat.

Step 2: A terminal window will start the backend and frontend servers.

Step 3: Your browser opens automatically with the beta version showing your live data and changes.

Step 4: Test your features. When finished, simply close the terminal window.

[DEVELOPER] start_pos.bat

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\start_pos.bat

When to Use: When running command-line debugging or technical inspections.

What it Does: Lightweight command-line script that boots uvicorn and Vite concurrently.

How to Use / Steps:

Step 1: Double-click start_pos.bat.

Step 2: Inspect console log outputs for API requests, barcode scans, and receipt print jobs.

2. Database Management & Diagnostic Doctors

[DATABASE SETUP] Setup_PostgreSQL_Database.bat (and setup_database.ps1)

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\Setup_PostgreSQL_Database.bat

When to Use: On initial software setup, when moving to a brand new laptop, or if PostgreSQL needs to be re-installed.

What it Does: Interactive setup engine. It asks for your desired Database Name, Username, Password, and Port (or accepts defaults on Enter). If PostgreSQL is not installed, it downloads and installs PostgreSQL 16 silently. It creates the database with UTF-8 encoding and writes .env configurations automatically.

How to Use / Steps:

Step 1: Right-click Setup_PostgreSQL_Database.bat and select 'Run as administrator'.

Step 2: The interactive prompt opens. Press [ENTER] to accept recommended defaults (dollytoyskidsweare, postgres, somesh123, 5432), or enter custom values.

Step 3: Press [ENTER] on the confirmation screen [Y/N] to proceed.

Step 4: When the green success box appears, press Enter to finish.

[DB SWITCHER] **Change_Database_Connection.bat (and change_database_connection.ps1)**

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\Change_Database_Connection.bat

When to Use: When you want Dolly POS to point to a different database (e.g., 'database1', 'dollytoyskidswear', 'testdb') or updated PostgreSQL password in 5 seconds.

What it Does: Connects to PostgreSQL, checks if the target database exists (creates it safely if missing), and writes the new connection parameters to .env files across all system locations. It never deletes existing tables or records.

How to Use / Steps:

Step 1: Double-click Change_Database_Connection.bat.

Step 2: It displays your current connection. Type your target database name (e.g. database1) and press Enter.

Step 3: Press Enter on Username, Password, and Port to keep existing values (or enter new ones).

Step 4: Press [ENTER] to confirm. Dolly POS is now switched to the target database in 5 seconds!

[SERVICE DOCTOR] **Heal_and_Restart_PostgreSQL.bat (and heal_and_restart_postgresql.ps1)**

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\Heal_and_Restart_PostgreSQL.bat

When to Use: If your laptop battery died, power went out abruptly, or PostgreSQL service stopped running.

What it Does: The 1-click diagnostic doctor. It terminates hung zombie processes holding port 5432, removes stale 'postmaster.pid' lock files caused by abrupt power loss, sets service to 'Automatic' startup, restarts the service, and verifies connection.

How to Use / Steps:

Step 1: Double-click Heal_and_Restart_PostgreSQL.bat.

Step 2: If Windows asks for Administrator permission, click 'Yes'.

Step 3: The doctor runs in 3 seconds and outputs a green success box.

Step 4: Launch Dolly POS and resume live billing.

[DATA SAFETY] **backup_db.bat (and scripts\backup_db.ps1)**

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\backup_db.bat

When to Use: Before making major system upgrades or for manual weekly data archives.

What it Does: Executes pg_dump to produce a timestamped SQL snapshot of the active PostgreSQL database in %USERPROFILE%\DollyPOS_Backups.

How to Use / Steps:

Step 1: Double-click backup_db.bat.

Step 2: A green confirmation shows the backup file path and size (e.g., dollytoyskidswear_20260926.sql).

[MAINTENANCE] **scripts\database_optimizer_and_vacuum.ps1**

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\scripts\database_optimizer_and_vacuum.ps1

When to Use: Once a month or after high-volume festive seasons (Diwali, Eid, New Year) to keep the database running at peak microsecond speed.

What it Does: Runs PostgreSQL VACUUM ANALYZE, re-indexes B-Tree and Trigram barcode search indexes, and cleans up old dead transaction tuples.

How to Use / Steps:

Step 1: Right-click database_optimizer_and_vacuum.ps1 → 'Run with PowerShell'.

Step 2: The script executes in 2 seconds and reports optimized table counts.

3. Packaging, Compiling & Update Tools

[PIPELINE COMPILER] Update_Production_App.bat

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\Update_Production_App.bat

When to Use: After approving changes in the Beta Launcher, when you want those changes pushed into the real desktop app (DollyPOS.exe) and setup installer.

What it Does: Runs the complete 3-step automated compilation pipeline: 1) Builds React frontend, 2) Compiles DollyPOS.exe via PyInstaller, 3) Packages single-file installer DollyPOS_Setup_v1.0.0.exe.

How to Use / Steps:

Step 1: Double-click Update_Production_App.bat.

Step 2: Wait ~45 seconds while it compiles.

Step 3: When finished, your desktop icon and installer are 100% updated with all new features.

[SHORTCUT INSTALLER] DollyPOS_Setup_Installer.bat

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\DollyPOS_Setup_Installer.bat

When to Use: To quickly create desktop shortcuts, Start Menu entries, and register Dolly POS in Windows Control Panel.

What it Does: Configures Windows desktop shortcuts pointing to dist_app\DollyPOS\DollyPOS.exe with proper working directories and icon associations.

How to Use / Steps:

Step 1: Double-click DollyPOS_Setup_Installer.bat.

Step 2: Desktop shortcuts and Start Menu icons are instantly created.

[STANDALONE INSTALLER] dist_installer\DollyPOS_Setup_v1.0.0.exe

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\dist_installer\DollyPOS_Setup_v1.0.0.exe

When to Use: When installing Dolly POS on a brand new laptop in 1 single file.

What it Does: Self-extracting GUI installation wizard. Extracts Dolly POS into %LOCALAPPDATA%\DollyPOS, configures .env settings, creates shortcuts, and registers in Windows Control Panel.

How to Use / Steps:

Step 1: Double-click DollyPOS_Setup_v1.0.0.exe.

Step 2: The pink installation progress bar runs to 100%.

Step 3: Click 'Launch Dolly POS Now'.

[UNINSTALLER] installer\Uninstall-DollyPOS.bat

Absolute Path: c:\Users\SomeshBang\Desktop\Antigravity DollyPos\installer\Uninstall-DollyPOS.bat

When to Use: If you ever need to cleanly remove the application binaries from a computer without losing your store data.

What it Does: Removes desktop shortcuts, Start Menu entries, and application binaries from %LOCALAPPDATA%\DollyPOS while preserving your store database and backups in DollyPOS_Backups.

How to Use / Steps:

Step 1: Run via Windows Control Panel 'Add or Remove Programs' or double-click Uninstall-DollyPOS.bat.

Step 2: Type 'Y' to confirm.

4. Emergency Offline Billing & Sync Engine

How Emergency Offline Billing Works: If PostgreSQL is stopped for any reason, Dolly POS switches automatically to local SQLite (dollypos_local.db). Cashiers continue billing, printing receipts, and adding customers with zero interruption.

Automatic Synchronization (offline_sync_service.py): When PostgreSQL comes back online, Dolly POS automatically scans for dollypos_local.db:

1. Transferred all offline bills, invoice items, and payments into PostgreSQL.
2. Creates new customers and updates Khata credit balances.
3. Automatically deducts sold product inventory in PostgreSQL.
4. Transferred any offline expenses and returns.
5. Archives the offline database to dollypos_local.db.synced_timestamp.bak so records are never duplicated while keeping full audit history.

Manual Trigger: You can also trigger offline sync anytime by clicking '**Sync Offline Bills**' in Settings → Database Backup & Safety or calling POST /api/v1/backup/sync-offline-sqlite.

5. Master Documentation & PDF Manuals Library

Document Name	Location Path	Contents & Description
Master File & Utility Dictionary	docs\Dolly_POS_Master_File_and_Utility_Dictionary.pdf	This document! Complete operational reference for every tool, script, launcher, and diagnostic utility.

Master Handbook & Troubleshooting Guide	docs\Dolly_POS_Master_Handbook_and_Troubleshooting_Guide.pdf	Daily store operations, cashier training, WhatsApp marketing, barcode generation, P&L; reports, and FAQ.
Master Technical Handoff & Maintenance Manual	docs\Dolly_POS_Master_Technical_Handoff_and_AI_Maintenance_Manual.pdf	Deep architecture manual for software engineers and AI assistants with API endpoints, schemas, and build systems.
Database Schema & ER Diagrams	docs\Dolly_POS_Database_Schema_and_ER_Diagrams.pdf	Full visual Entity-Relationship diagrams and column-by-column descriptions of all 18 database tables.
User Manual	docs\Dolly_POS_User_Manual.pdf	Simplified 1-page quick guide for cashiers and counter staff.
TSC TE244 Barcode Printer Guide	docs\TSC_TE244_Thermal_Printer_Setup_Guide.pdf	Configuration guide for thermal barcode sticker calibration, 50x25mm labels, and TSPL printer drivers.
New Laptop Setup Guide	dist_installer\NEW_LAPTOP_SETUP_README.md	Step-by-step instructions for deploying Dolly POS on a new computer.